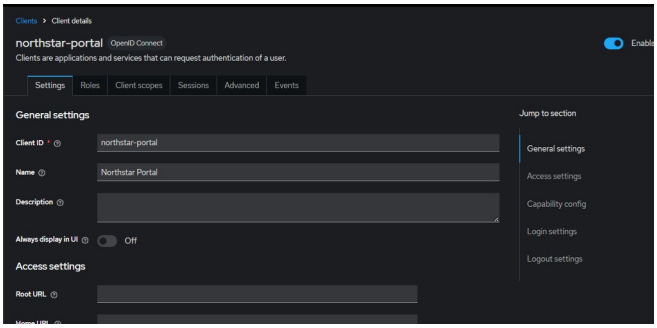
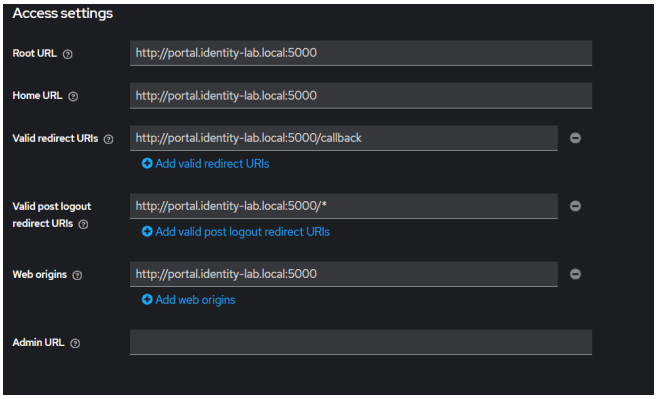
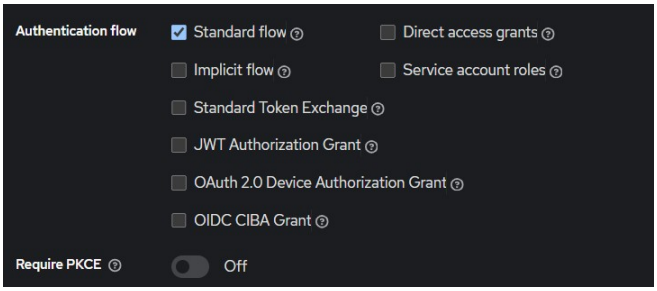
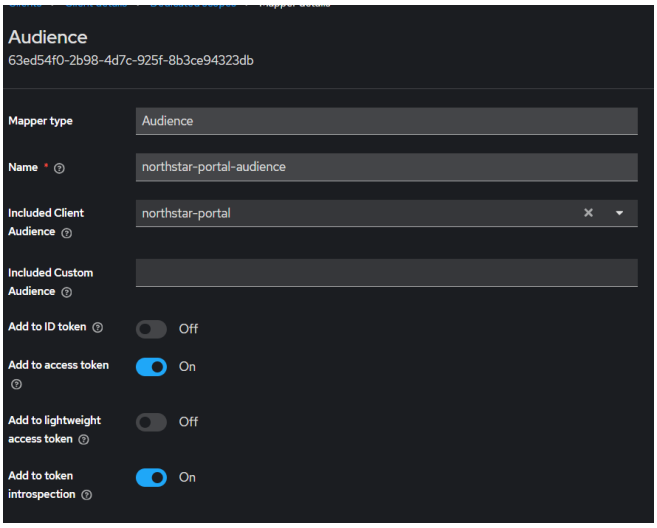
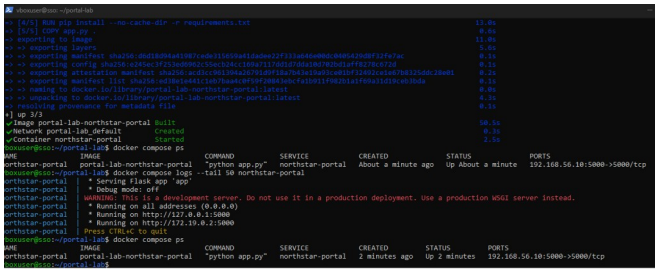
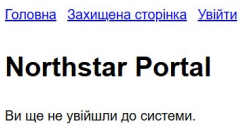
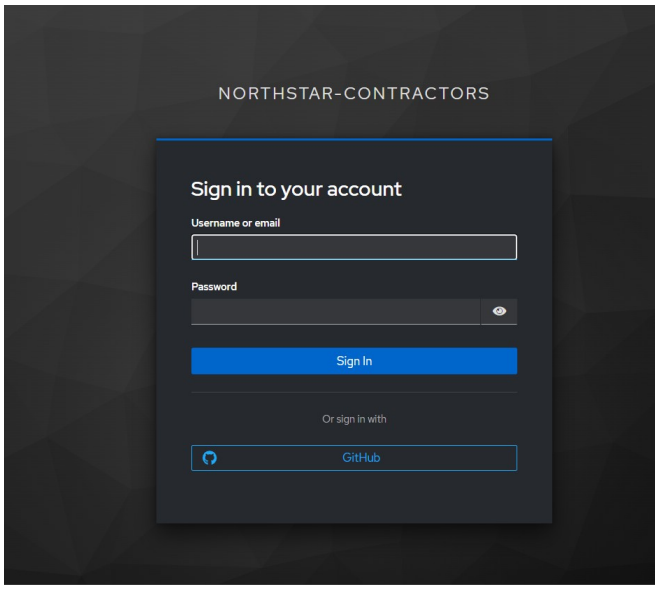
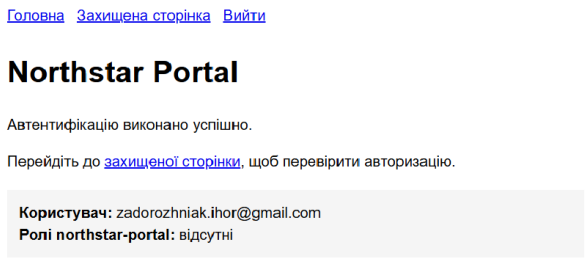
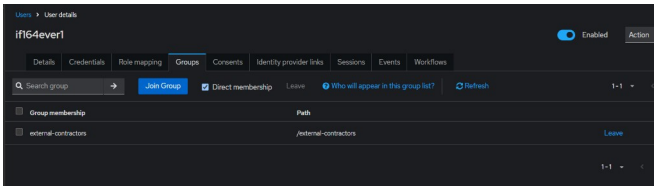
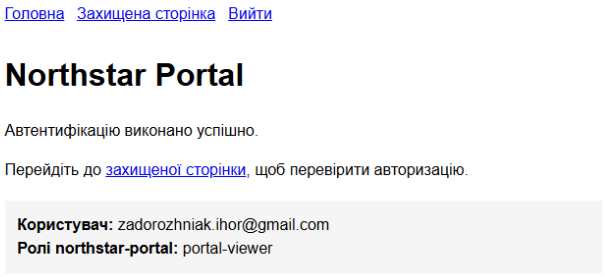
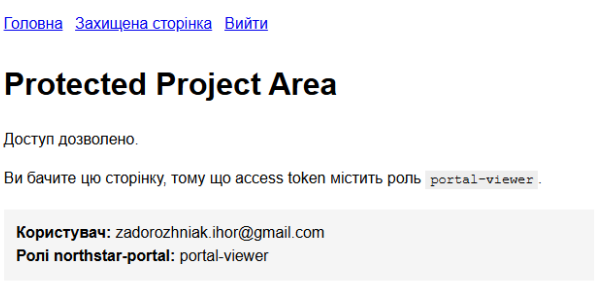
1. Client Authentication Configuration Client authentication was enabled for the `northstar-portal` client. This allows the application to authenticate using a client secret when exchanging an authorization code for tokens.	
2. Application URLs and Redirect URI The application URLs were configured for `http://portal.identity-lab.local:5000`. The exact redirect URI `/callback` was registered to receive the authentication response from Keycloak.	
3. Standard Flow Configuration Standard flow was enabled for the client. This allows the portal to use the Authorization Code Flow for user login.	
4. Audience Mapper Configuration An Audience mapper was created to include `northstar-portal` in the access token's audience claim. This allows the application to verify that the token is intended for this client.	
5. Application Container Deployment	

The Northstar Portal application was built and started using Docker Compose. The container was confirmed to be running with port 5000 published on the Ubuntu VM’s host-only address.	
6. Portal Home Page The portal was opened at `http://portal.identity-lab.local:5000`. The home page displayed a login link for authentication through Keycloak.	
7. Keycloak Login Page Selecting the portal’s login link redirected the browser to Keycloak. GitHub was available as an external identity provider.	
8. Successful Authentication Without a Role The user successfully authenticated through GitHub and returned to the portal. The user information panel showed that no roles were assigned for `northstar-portal`.	
10. Group Membership Assignment	

The user was added to the `external-contractors` group in Keycloak. Through this membership, the user inherited the group's `portal-viewer` client role.	
11. New Login With the Required Role The user logged out and signed in again to obtain a new access token. The portal displayed `portal-viewer` in the user's client role list.	
12. Successful Access to the Protected Page The application allowed the user to open the Protected Project Area. Access was granted because the verified access token contained the `portal-viewer` role.	
13. Access Revocation Verification The user was removed from the `external-contractors` group and signed in again. The new token no longer contained `portal-viewer`, so the application denied access to the protected page.	